



Instructor: Date:

Learner: Regional Branch:

Program: Center:

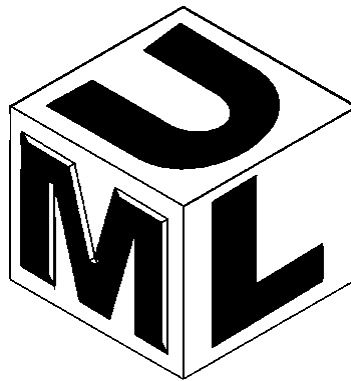
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Let's Start!

What do you know about Unified Modeling Language?



Learning Outcome:

Describe common activities at work using short sentences in familiar spoken and written situations.

Interpret the requirements report to determine the technological needs in managing information, based on rules and protocols established by the company.

Can Do Statements:

- I can explain the concept of UML.
- I can give additional information about UML.
- I can ask questions about UML.

Ask your partner:

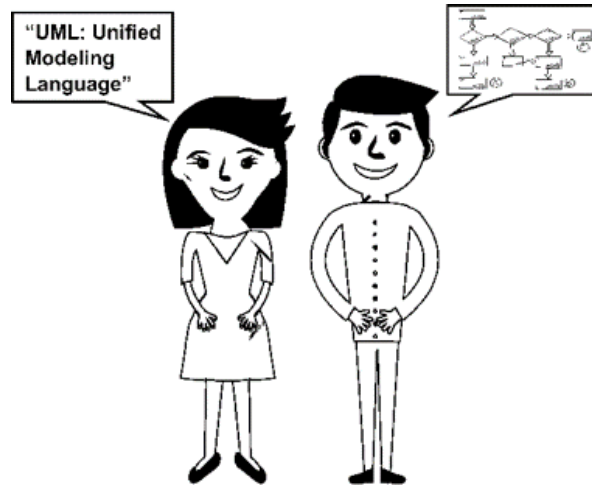
- What programming languages do you know? Name some.
- Do you know what the difference between programming language and modeling language is? Explain.





Let's Explore!

Peter couldn't make it to programming class yesterday.



Instructions: Read the text below.

- Peter:** I couldn't make it to class yesterday. Do you know if we have anything for next session?
- Liz:** Yes, for next class, Mr. Collins wants us to make a presentation about UML. So, I advise you to be prepared.
- Peter:** Really? Can you help me catch up? What do you know about UML?
- Liz:** Well, UML means Unified Modeling Language. It permits programmers to define, visualize, build, and describe software tools.
- Peter:** I see. So, UML enables users to have a standardized way to perceive the design of a system. Right?
- Liz:** Exactly. UML uses meta object facility and graphic notation. This is a key factor in object-oriented programming because it allows programmers to create visual models of systems.
- Peter:** OK. But, is UML useful during the software development cycle?
- Liz:** Absolutely. If you need to develop a software project, you can use UML during the whole cycle. This modeling language is able to produce entire applications. Besides, UML executables can be deployed in different types of platforms.
- Peter:** Wow! That seems pretty cool. And, what kind of elements does UML offer?
- Liz:** Well, UML includes programming language statements, actors, activities, business processes, and logical and reusable software components.
- Peter:** I understand. And, do all UML diagrams work the same way?
- Liz:** Not really. If you use UML diagrams, you need to know that they are divided into two categories. In the first type, you can find diagrams that can represent structural information. This type of diagrams allows programmers to document the architecture of software systems. The second type of diagrams represent general types of behavior, we use them to describe the functionality of systems.



Let's Practice!

UML is based on a meta-object facility.

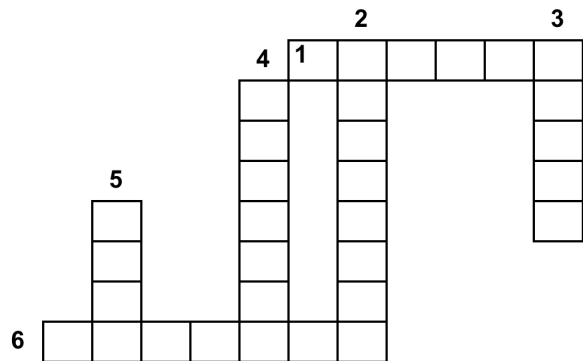
Instructions: Read the clues. Complete the crossword.

Across

1. Some UML diagrams **p** _ _ _ _ _ programmers to document the architecture of software systems.
6. Mr. Collins **a** _ _ _ _ _ us to use UML because different platforms can read UML executables.

Down

2. UML **e** _ _ _ _ _ users to have a standardized way to perceive the design of a system.
3. Liz **t** _ _ _ _ Peter to review all the elements that UML include such as programming language statements, actors, and business processes.
4. UML is key in object-oriented programming because it **a** _ _ _ _ _ programmers to create visual models of systems.
5. We **n** _ _ _ some other UML diagrams to describe the functionality of systems.



Let's Practice!

UML is a very precise modeling language.

Instructions: Use the words from the Word Bank to complete each sentence.

Word Bank

using / uses / use / needs / to need / need to / require / required / requires

1. If you _____ UML, you don't have to program anything.
2. If you _____ show a process to a client, you use UML to represent it.
3. If you _____ UML, it's easier to show the sections of a process.





Extension Activity

What does UML mean?

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about the concept and applications of UML.
- Identify key facts and summarize them.
- Create 5 questions and answers about UML.
- Create a brochure to explain UML using the questions and answers you created.
- Share your work with your classmates.

Self-Assessment

Yes

No

- | | | |
|--|-----------------------|-----------------------|
| 1. I look for more information about the concept and applications of UML. | ☐ | ☐ |
| 2. I identify and summarize key information. | ☐ | ☐ |
| 3. I create 5 questions and answers about UML. | ☐ | ☐ |
| 4. I create a brochure to explain UML using the questions and answers you created. | ☐ | ☐ |
| 5. Share your work with your classmates. | ☐ | ☐ |



Target Words

- **standardized** (adjective): when something has similar features with regard to some established requirements or rules
e.g. UML enables users to have a standardized way to perceive the design of a system.
- **notation** (noun): a set of symbols to represent data
e.g. UML uses meta object facility and graphic notation.
- **executable** (noun): a file that has the proper format to be read
e.g. Besides, UML executables can be deployed in different types of platforms.
- **reusable** (adjective): when something can be used repeatedly
e.g. UML includes programming language statements, actors, activities, business processes, and logical and reusable software components.
- **meta object facility** (noun): a standard management group that provides a set of interfaces to manipulate systems
e.g. UML uses meta object facility and graphic notation.



Learning Points

Verbs + Objects + Infinitives.

We use the structure **verb + object + to infinitive** when we want to make a request, give an advice or a warning.

There are some verbs that always require to follow the pattern following in:

Subject + **verb + object + to infinitive** + complement

Examples:

- *She **advised me to review** more information about UML.*
- *They **want the company to develop** their software project.*

Remember that the object of the sentence is the person or the thing that receives the action. In the examples above the objects are *me* and *the company*.

In the following chart, you can find some of the verbs that use this structure:

Examples of Verbs Followed by an Object and the to-Infinitive			
advise	command	need	tell
allow	enable	permit	want
ask	expect	prepare	warn
cause	instruct	remind	would

Examples

- *Mr. Collins wants us to make a presentation about UML. So, I advise you to be prepared.*
- *It permits programmers to define, visualize, build, and describe software tools.*





Learning Points

Versions of UML.

UML is a standard language for modeling software systems.

It has different versions.

UML emerged the mid-1990 thanks to the development of object-oriented methods developed in the late 80's and early 90's. This modeling language integrates the notions of the Booch method, the object-modeling technique, and object-oriented software engineering.

There are two main versions of UML, 1.x and 2.x, each with multiple sub versions. In 1996, an international consortium called UML Partners proposed the first draft of this language to the OMG (Object Management Group). Then, a group of experts got together in order to define the language constructs and specifications to meet all the standardization purposes. This version, UML 1x, was finally released in November 1997.

After different versions of UML 2.X, such as UML 2.0, 2.2 and 2.4, the latest version, UML 2.5, appeared in 2015. These versions have common specifications and improvements in terms of Superstructure, Infrastructure, the Object Constraint Language, and UML Diagram Interchange.